Experiment no. 1

Aim: Demonstration Steps involved in EDA

Tools: Google Colab, Kaggle

Theory: Exploratory Data Analysis (EDA) is a systematic approach used to understand, summarize, and analyze datasets before applying advanced statistical or machine learning techniques. The main objective of EDA is to gain insights into the data by examining its structure, identifying patterns, detecting missing values, and understanding relationships between variables through statistical measures and visualizations.

In this experiment, EDA is performed on a state-wise inflation index dataset of India, which contains information such as sector, year, and inflation index values for different Indian states and union territories. The dataset includes both categorical variables (Sector, Name) and numerical variables (Year and state-wise inflation indices).

The experiment begins by **studying the data structure**, which includes determining the number of rows and columns, identifying data types, and understanding the overall format of the dataset. This step ensures familiarity with the dataset and helps in selecting appropriate analysis methods.

Next, a **statistical summary** is generated to compute key descriptive statistics such as mean, median, minimum, maximum, and standard deviation for numerical columns, and frequency counts for categorical columns. This helps in understanding the central tendency and dispersion of the inflation data.

Univariate analysis is then performed to study individual variables separately using techniques such as histograms, bar charts, and box plots. This analysis helps in understanding the distribution and variability of inflation values for each state and across years.

Bivariate analysis is carried out to examine relationships between two variables, such as year-wise inflation changes for a particular state or comparison of inflation between two states. Scatter plots and box plots are used to identify trends, correlations, and variations.

Multivariate analysis is performed to analyze relationships among multiple state-wise inflation variables simultaneously. Correlation matrices and heatmaps are used to identify similarities and dependencies between inflation trends of different states.

The experiment also includes **missing value analysis**, where null or missing values are identified and quantified. This step is important to ensure data quality and reliability of the analysis.

Thus, through systematic EDA techniques and visualizations, meaningful insights are extracted from the inflation dataset, enabling better understanding and interpretation of state-wise inflation trends in India.

Code and Output:

Step 1: Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 2: Add Data Set

```
df = pd.read_csv('/content/Statewise_General_Index_Upto_Feb24.csv')
```

Step 3: Find Data structure

```
# Shape of dataset
df.shape
# Column names
df.columns
# Data types and non-null counts
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 471 entries, 0 to 470
Data columns (total 39 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Sector                  471 non-null    object
 1   Year                    471 non-null    int64
 2   Name                    471 non-null    object
 3   Andhra Pradesh          461 non-null    float64
 4   Arunachal Pradesh       81 non-null     object
 5   Assam                   462 non-null    float64
 6   Bihar                   117 non-null    float64
 7   Chattisgarh             462 non-null    float64
 8   Delhi                   117 non-null    float64
 9   Goa                     462 non-null    float64
 10  Gujarat                 462 non-null    float64
 11  Haryana                 462 non-null    float64
 12  Himachal Pradesh        456 non-null    object
 13  Jharkhand               462 non-null    float64
 14  Karnataka               462 non-null    float64
 15  Kerala                  462 non-null    float64
 16  Madhya Pradesh          462 non-null    float64
 17  Maharashtra             462 non-null    float64
 18  Manipur                 462 non-null    float64
 19  Meghalaya               462 non-null    float64
 20  Mizoram                 462 non-null    float64
 21  Nagaland                117 non-null    float64
 22  Orissa                  462 non-null    float64
 23  Punjab                  462 non-null    float64
 24  Rajasthan               117 non-null    float64
 25  Sikkim                  117 non-null    float64
 26  Tamil Nadu              462 non-null    float64
 27  Telangana               315 non-null    float64
 28  Tripura                 456 non-null    float64
 29  Uttar Pradesh           462 non-null    float64
 30  Uttarakhand             117 non-null    float64
 31  West Bengal             456 non-null    float64
 32  Andaman and Nicobar     462 non-null    float64
 33  Chandigarh              462 non-null    float64
 34  Dadra and Nagar Haveli  462 non-null    float64
 35  Daman and Diu           462 non-null    float64
 36  Jammu and Kashmir       462 non-null    float64
 37  Lakshadweep             462 non-null    float64
 38  Puducherry              462 non-null    float64
dtypes: float64(34), int64(1), object(4)
```

Step 4: Statistical Summary

```
df.describe()
df['Andhra Pradesh'].describe()
```

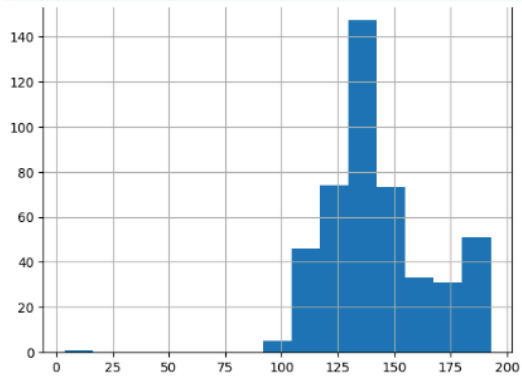
	Year	Andhra Pradesh	Assam	Bihar	Chattisgarh	Delhi	Goa	Gujarat	Haryana	Jharkhand	...
count	471.000000	461.000000	462.000000	117.000000	462.000000	117.000000	462.000000	462.000000	462.000000	462.000000	...
mean	2017.076433	143.001996	141.581104	169.275385	140.462035	161.055897	140.709740	139.270216	137.928528	141.844697	...
std	3.812423	23.417518	23.365581	18.252225	20.112276	15.630042	21.362898	19.975204	21.237620	21.835543	...
min	2011.000000	3.640000	0.790000	1.620000	1.220000	5.640000	0.250000	6.820000	3.350000	1.390000	...
25%	2014.000000	128.300000	125.100000	163.600000	125.700000	158.500000	125.450000	125.325000	123.075000	126.425000	...
50%	2017.000000	138.800000	137.250000	170.900000	139.050000	164.000000	139.200000	136.150000	134.350000	139.500000	...
75%	2020.000000	154.900000	157.200000	178.600000	154.900000	166.400000	160.275000	154.350000	150.900000	157.100000	...
max	2024.000000	192.800000	189.800000	189.300000	181.400000	172.500000	176.600000	184.900000	189.500000	188.700000	...

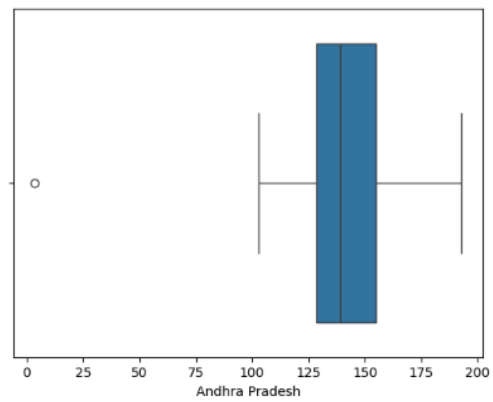
8 rows × 35 columns

	Andhra Pradesh
count	461.000000
mean	143.001996
std	23.417518
min	3.640000
25%	128.300000
50%	138.800000
75%	154.900000
max	192.800000

Step 5: Univariate Data

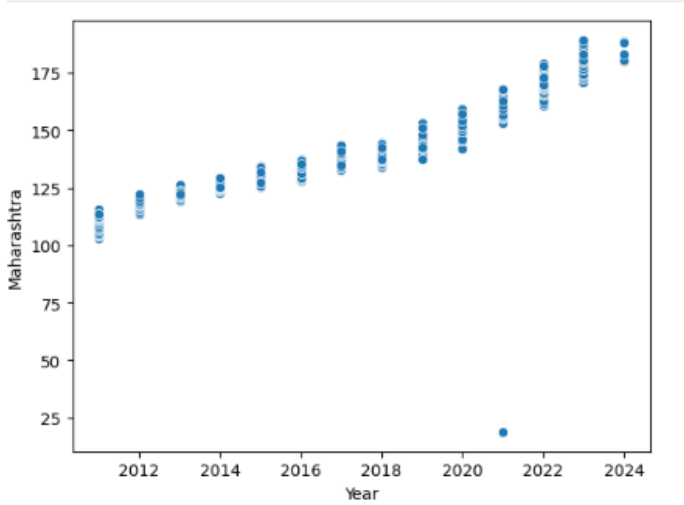
```
df['Andhra Pradesh'].hist(bins=15)
plt.show()

sns.boxplot(x=df['Andhra Pradesh'])
plt.show()
```





Step 6: Bivariate data

```
sns.scatterplot(x='Year', y='Maharashtra', data=df)
plt.show()
```



Step 7: Multivariate Data

```
selected_states = ['Andhra Pradesh','Gujarat','Maharashtra','Tamil
Nadu','West Bengal']
filtered = df_melted[df_melted['State'].isin(selected_states)]
plt.figure(figsize=(12,6))
sns.lineplot(
    data=filtered,
    x='Year',
    y='Inflation',
    hue='State'
)
plt.title('Inflation Trend of Selected States')
plt.show()
```


```
selected_states = ['Andhra Pradesh','Gujarat','Maharashtra','Tamil
Nadu','West Bengal']
filtered = df_melted[df_melted['State'].isin(selected_states)]
plt.figure(figsize=(12,6))
sns.lineplot(
    data=filtered,
    x='Year',
    y='Inflation',
    hue='State'
)
plt.title('Inflation Trend of Selected States')
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```


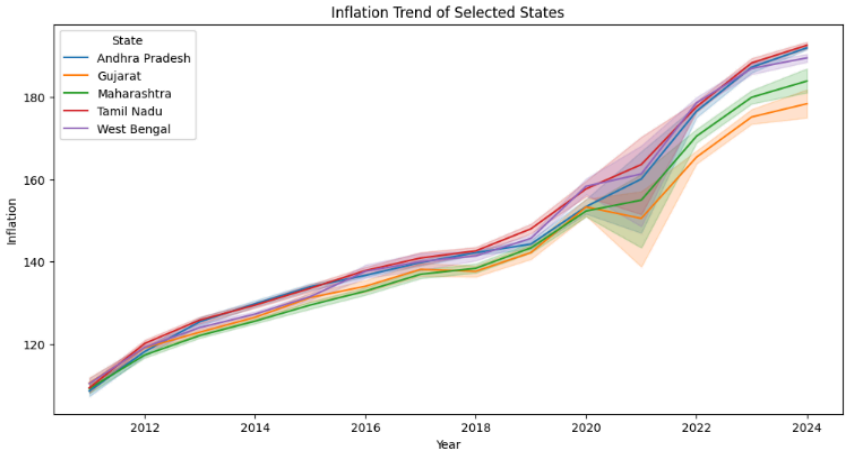

Conclusion:

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